POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Object Oriented Programming in C++ (New) | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What makes object-oriented programming better than procedure-oriented programming? Justify your answer with features of object-oriented programming. 7
   b) List out the advantages of adopting RDD. Draw CRC cards for Library Management system. 8
2. a) What are static data members and static functions? Explain with examples. 7
   b) What is constructor? Explain three types of constructor with example program. 8
3. a) Create a class Person with data members Name and age. Derive a class Employee from Person with its data member employee_id. Derive another class Manager from Employee with its data member department. Receive the data members from each class's member function and display the data of all classes from the class Manager. 8
   b) How composition differs from inheritance? Write a program to concatenate two strings (name and address of a person) using the concept of containership. 7

OR

During the time of hybrid inheritance when there is hierarchical inheritance at the upper level and multiple inheritance at lower level, ambiguity occurs due to duplication of data from multipath at the grand child class. How this kind of ambiguity is resolved? Explain with suitable example.

4. a) What is the major difference between compile time and runtime polymorphism? How runtime polymorphism can be achieved? Explain with example. 7
   b) What do you mean by type casting? Write a program to convert an object of Rectangle class into the object of Polar class by using type conversion routine. 8

OR

What are functions overloading and overriding? Explain with examples.

5. a) What are the advantages of function template? Create a function template to swap two integers, two floating point data and two characters. 7
   b) What is the Standard Template Library (STL)? Explain any one container of STL with example. 8
6. a) What is exception? Describe exception handling in C++ with example. 7
   b) Write a program that prompts the user to enter their name and age, write the entered data into a file named "userinfo.txt", read the details from "userinfo.txt" and display the data on the console. 8
7. Write short notes on: (Any two) 2×5
   a) Software reusability
   b) Dynamic Memory Allocation
   c) Computation as simulation

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